

Question description

Your task is to maintain an array of n values and efficiently process the following types of queries:

1. Increase the first value in range $[a,b]$ by 1, the second value by 2, the third value by 3, and so on.
2. Calculate the sum of values in range $[a,b]$.

Input

The first input line has two integers n and q : the size of the array and the number of queries.

The next line has n values $t_1, t_2, \dots, t_n$: the initial contents of the array.

Finally, there are q lines describing the queries. The format of each line is either "1 a b" or "2 a b".

Output

Print the answer to each sum query.

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq t_i \leq 10^6$
- $1 \leq a \leq b \leq n$

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

5 3
3 7 2 2 5
3 2 3
1 3 6
1 2 3

EXPECTED OUTPUT

9

Test Case 2

INPUT (STDIN)

5 3
4 2 3 1 7
2 1 5
1 1 5
2 1 5

EXPECTED OUTPUT

17
32

▼ Mandatory Test Cases

Test Case 1

KEYWORD

void build(int k,int
l,int r)

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

2

Test Case 2

TOKEN COUNT

800

Test Case 3

NLOC

71